beaverLacAreaMtoe8ha model summary

Contents

Overview

This document summarizes the results of a global sensitivity and uncertainty analysis for the **beaverLacAreaMtoe8ha** habitat suitability index (HSI) model for *Castor canadensis*. Metadata for the model is stored in the ecorest package in R.

The original documentation for this model can be found here[1].

Sub-model: **Lacustrine habitat with surface area more than or equal to 8 ha (20 acres)**

The beaverLacAreaMtoe8ha model is comprised of **8** variables and **2** components.

Variables:

Table 1. SIV variables included in the beaverLacAreaMtoe8ha model. Type indicates whether a variable is continuous or categorical and breakpoints indicates the number of distinct breakpoints in suitability graphs.

	Variable name	Type	Breakpoints
SIV1	can.closure.SIV	continuous	4
SIV2	trees.2.5.to.15.2cm.SIV	continuous	2
SIV3	shrub.crown.cov.SIV	continuous	4
SIV4	avg.h.shrub.can.SIV	continuous	3
SIV5	species.comp.SIV	categorical	3
SIV6	lac.surface.dominated.yellow.white.wtr.lilly.SIV	continuous	2
SIV8	avg.wtr.fluct.SIV	categorical	3
SIV9	shoreline.dev.factor.SIV	continuous	3

[1] https://ecolibrary.sec.usace.army.mil/resource/c720bcec-c944-47f9-988a-549266cea773

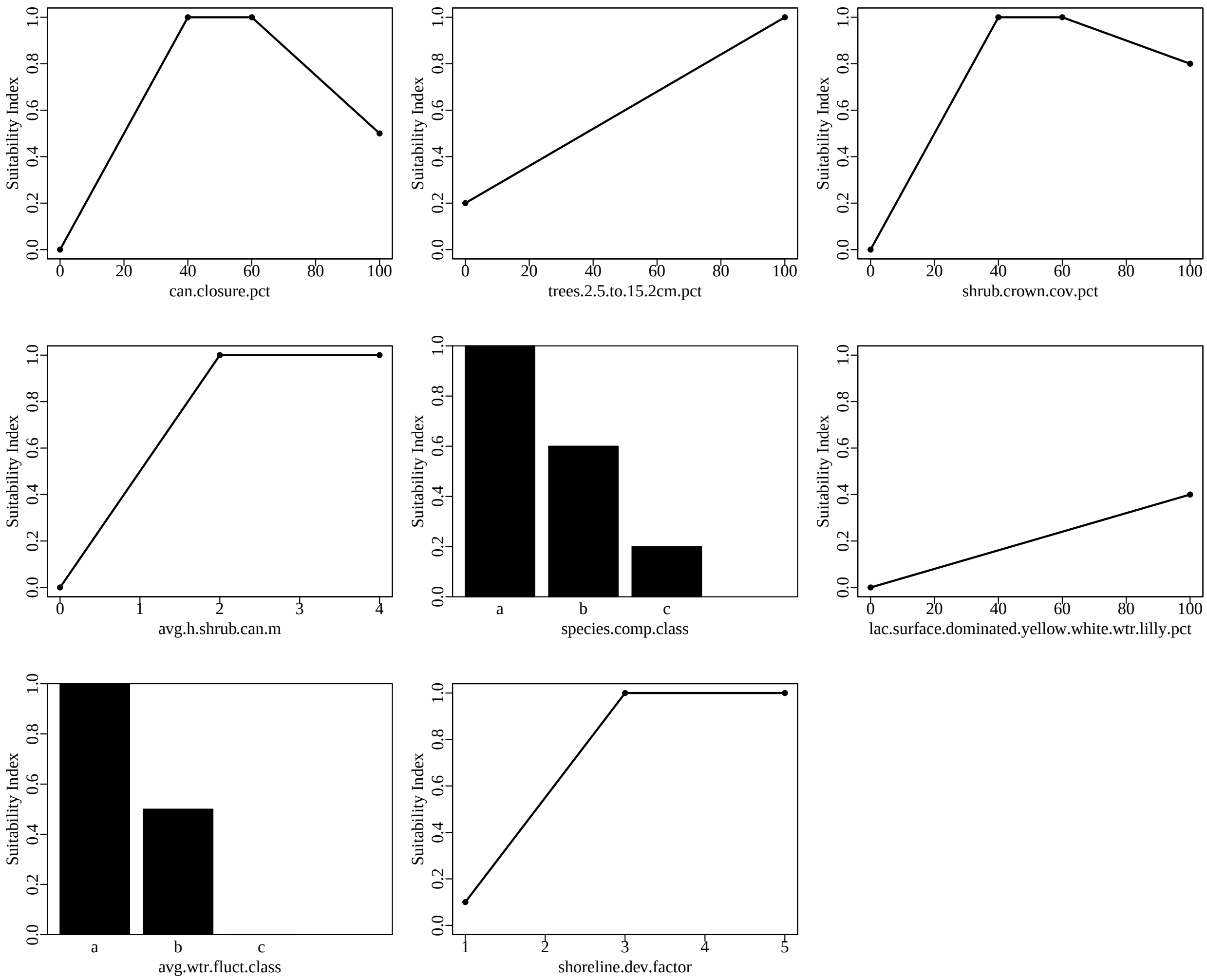

Figure 1. Suitability index graphs for variables included in the beaverLacAreaMtoe8ha model in ecorest.

Components:

Table 2. Components included in the beaverLacAreaMtoe8ha model in ecorest.

Component	Equation
Winter food component	`((((((SIV1*SIV2)^(1/2))*SIV5)^(1/2))+((((SIV3*SIV4)^(1/2))*SIV5)^(1/2))+0.5*(((((SIV1*SIV2)^(1/2))*SIV5)^(1/2))+((((SIV3*SIV4)^(1/2))*SIV5)^(1/2))))/1.5)+SIV6)`
Water component	`min(SIV8, SIV9)`

Model equation:

The equation to calculate an overall HSI index for the beaverLacAreaMtoe8ha model is:

`min(CWF, CW)`

According to our classification, this model's format is: **author-specified**

Global sensitivity and uncertainty analysis:

We ran global sensitivity and uncertainty analyses on the beaverLacAreaMtoe8ha model using the sensobol package in R (Puy et al. 2022). The following parameters were used for the sensobol analysis:

Table 3. Parameters and settings used for sensobol sensitivity and uncertainty analyses.

Parameter	Equation	Value
Number of input variables (M)	-	8
Base sample size (n)	-	102000
Number of model evaluations (N)	n*(M+2)	1020000
First order estimator	See Puy et al. (2022)	Saltelli
Total order estimator	See Puy et al. (2022)	Jansen
Number of bootstrap replications	-	1000
Sampling scheme	-	Quasi-random
Matrices	-	A, B, AB

We ran a sensitivity and uncertainty analysis for the beaverLacAreaMtoe8ha model using the original equation outlined in the documentation from Allen (1982) and using arithmetic mean, geometric mean, limiting factor, and multiplicative equations to contrast the results across different equation structures.

Model uncertainty

We ran the beaverLacAreaMtoe8ha model using 1020000 combinations of its SIV variables, which were sampled from a uniform distribution spanning the range of possible values listed in the beaverLacAreaMtoe8ha documentation. We limited the range of possible values for each parameter to the range in which the SIV values were greater than zero to prevent HSI score distributions with primarily zero values.

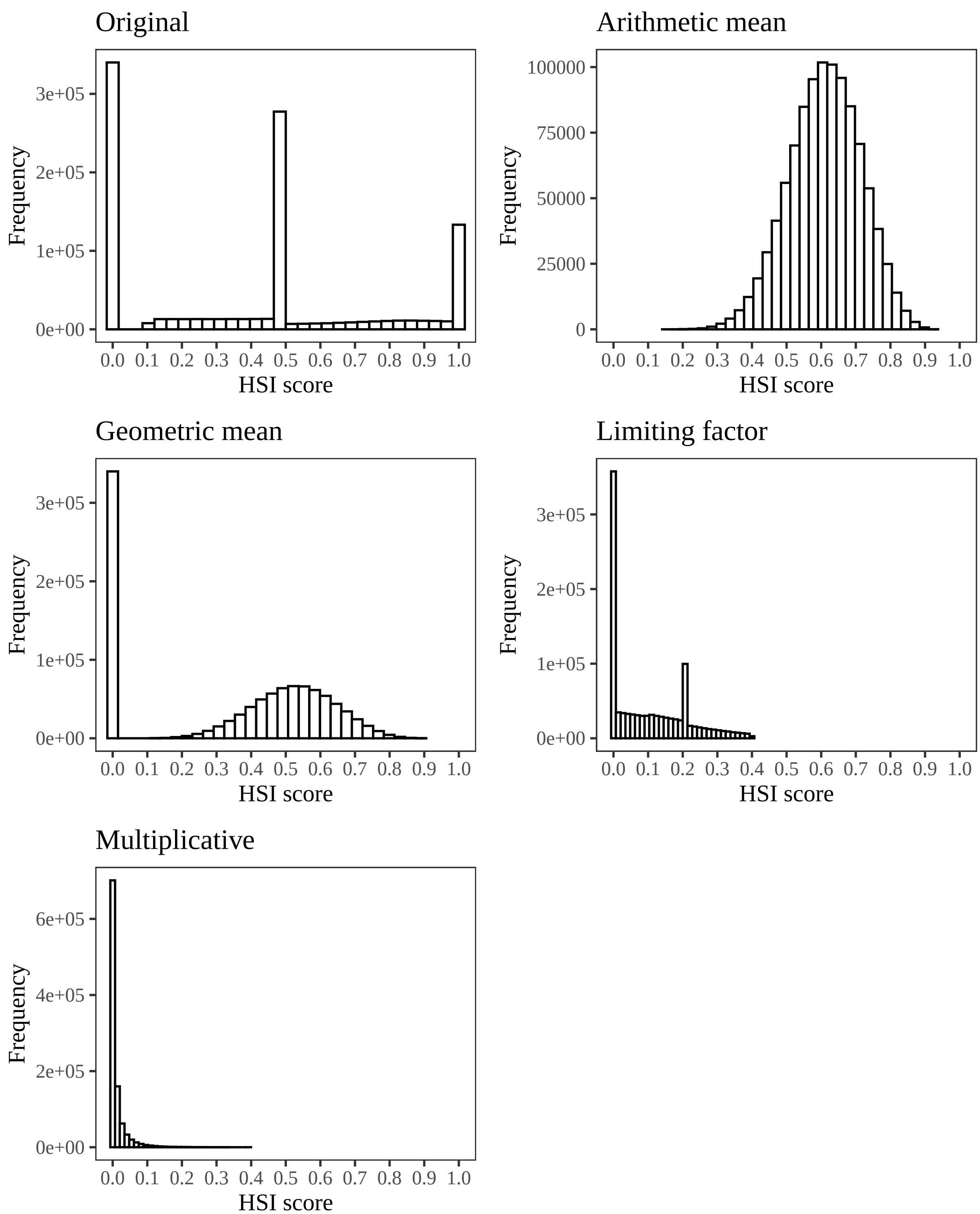

Figure 2. Empirical distributions of HSI scores for the beaverLacAreaMtoe8ha model using the original author-specified model equation from Allen (1982), and an arithmetic mean, geometric mean, limiting factor, and multiplicative structure incorporating all SIV variables. Note differences in the y axis.

We assumed a uniform distribution for all parameters because we evaluated all ecorest models in batch. Should you decide to run your own sensitivity analysis, this assumption should be evaluated independently for each parameter in the model.

Table 4. Quantiles from the empirical distribution of HSI scores for the original beaverLacAreaMtoe8ha model structure, an arithmetic mean equation, a geometric mean equation, a limiting factor equation, and a multiplicative equation structure.

	1%	**2.5%**	**5%**	**25%**	**50%**	**75%**	**95%**	**97.5%**	**99%**	**100%**
Original	0.00	0.0	0.00	0.00	0.50	0.54	1.00	1.00	1.00	1.00
Arithmetic	0.36	0.4	0.44	0.54	0.61	0.68	0.78	0.80	0.83	0.92
Geometric	0.00	0.0	0.00	0.00	0.44	0.57	0.70	0.73	0.77	0.89
Limiting	0.00	0.0	0.00	0.00	0.07	0.19	0.31	0.35	0.38	0.40
Multiplicative	0.00	0.0	0.00	0.00	0.00	0.01	0.06	0.08	0.12	0.40

Generally, models with outputs that span the entire range of possible outcomes (*i.e.*, 0-1) may be better at discriminating between different restoration actions or across habitat quality. The empirical distribution of HSI scores can also provide you with information about whether HSI scores for a given model tend to be skewed (*e.g.*, many low scores and few high scores, uniform probability of obtaining scores from 0-1, *etc.*)

Model sensitivity

Below are the results of the global sensitivity analysis for the beaverLacAreaMtoe8ha model using the original equation, an arithmetic mean, a geometric mean, a limiting factor, and a multiplicative model structure. The sensobol package uses variance-based sensitivity metrics, so the model's sensitivity to a given parameter is a measure of how much variance in the HSI score decreases in response to that parameter being fixed (Puy et al. 2022). For each parameter, the observed changes in the variance of the HSI score can be described with a first order sensitivity index (S_i) that accounts for the influence of a single parameter of interest on variance in HSI, or with a total order index (T_i) that accounts for the influence of a single parameter on its own and in combination with all other parameters (*i.e.*, interactions) (Puy et al. 2022). We can compare the 95% confidence intervals for the first and total order indices to a dummy parameter, which represents a parameter that has no influence on the variance in a model's output. While an uninfluential variable should theoretically have an S_i and T_i of zero, small approximation errors can lead variables to have a non-zero influence on a model's output (Puy et al. 2022). If the confidence interval of the S_i and T_i index for a given parameter overlaps the confidence interval of the dummy parameter, we can deduce that the parameter has a negligible effect on variance in HSI scores, both on its own and in combination with all other variables.

To ensure that the results of our sensitivity analysis converged, we determined whether the range of the 95% confidence interval for each sensitivity index was less than or equal to 0.05 (Sarrazin et al., 2016). Models that did not converge should be interpreted with caution.

Table 5. Model convergence records for the original beaverLacAreaMtoe8ha model structure, an arithmetic mean equation, a geometric mean equation, a limiting factor equation, and a multiplicative equation structure.

	Did the model converge? (T/F)
Original	TRUE
Arithmetic	TRUE
Geometric	TRUE
Limiting	TRUE
Multiplicative	TRUE

Original model

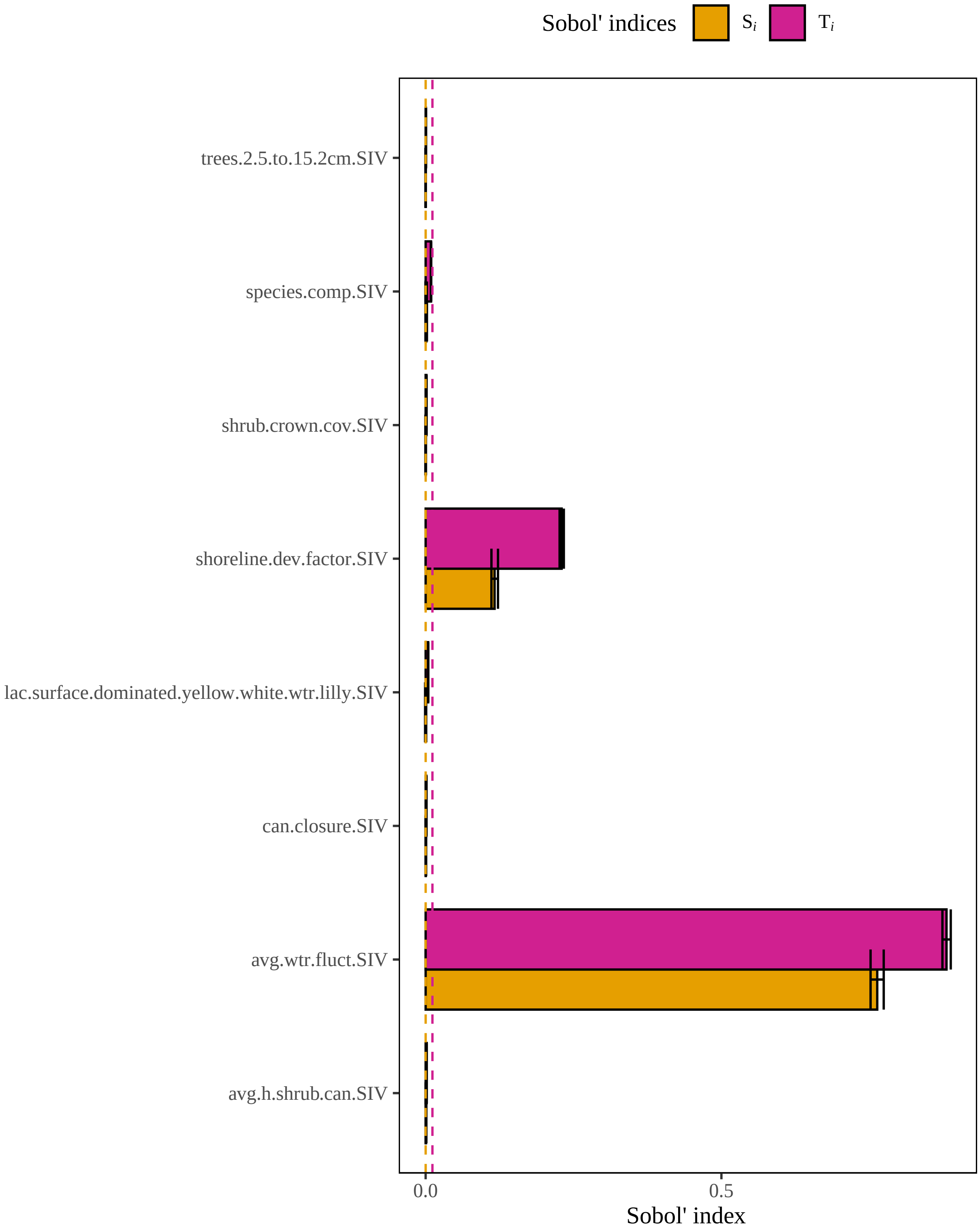

Figure 3. Results of a sensitivity analysis for the beaverLacAreaMtoe8ha model based on the original author-specified model outlined in Allen (1982). Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Arithmetic mean model

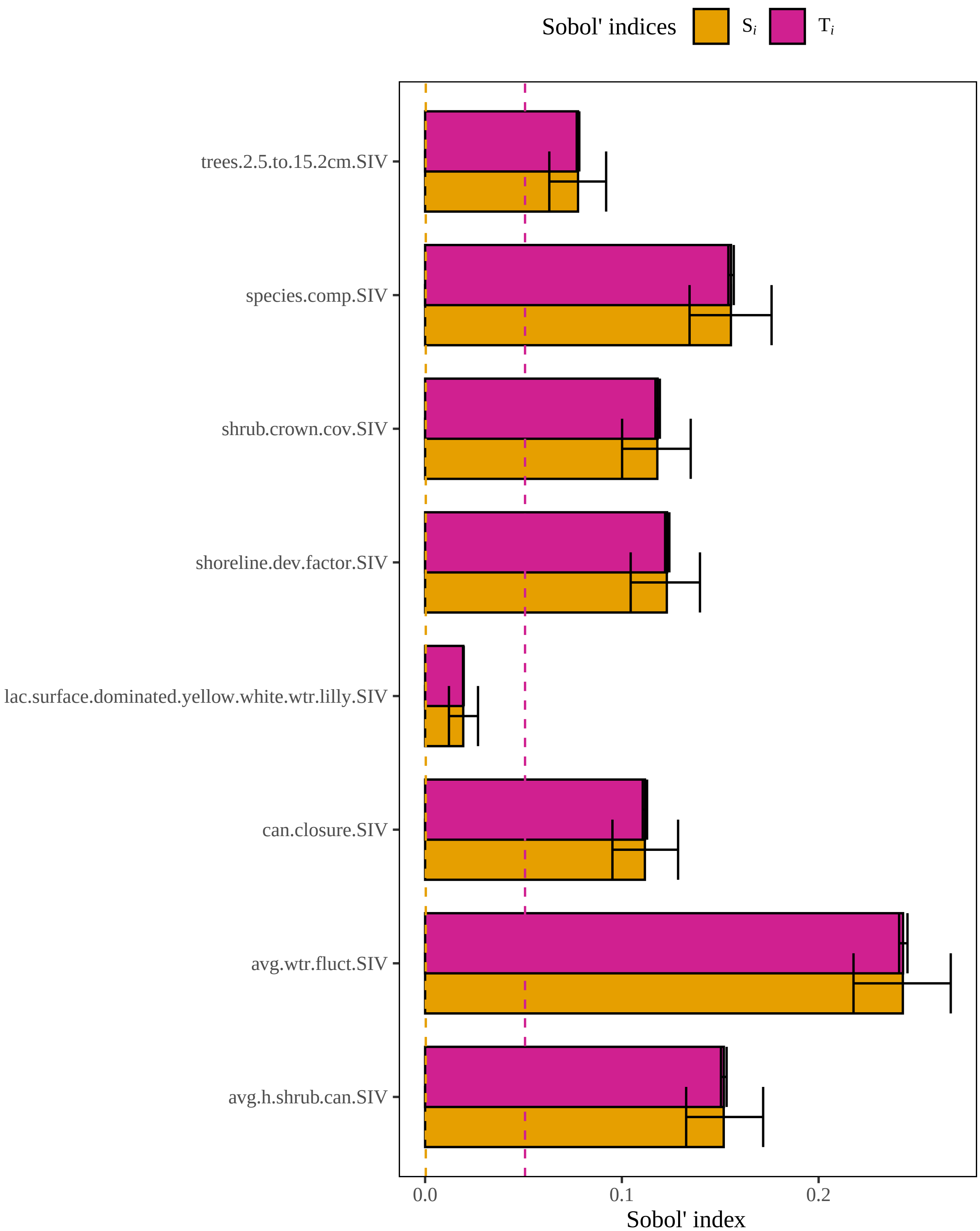

Figure 4. Results of a sensitivity analysis for the beaverLacAreaMtoe8ha model based on an arithmetic mean structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Geometric mean model

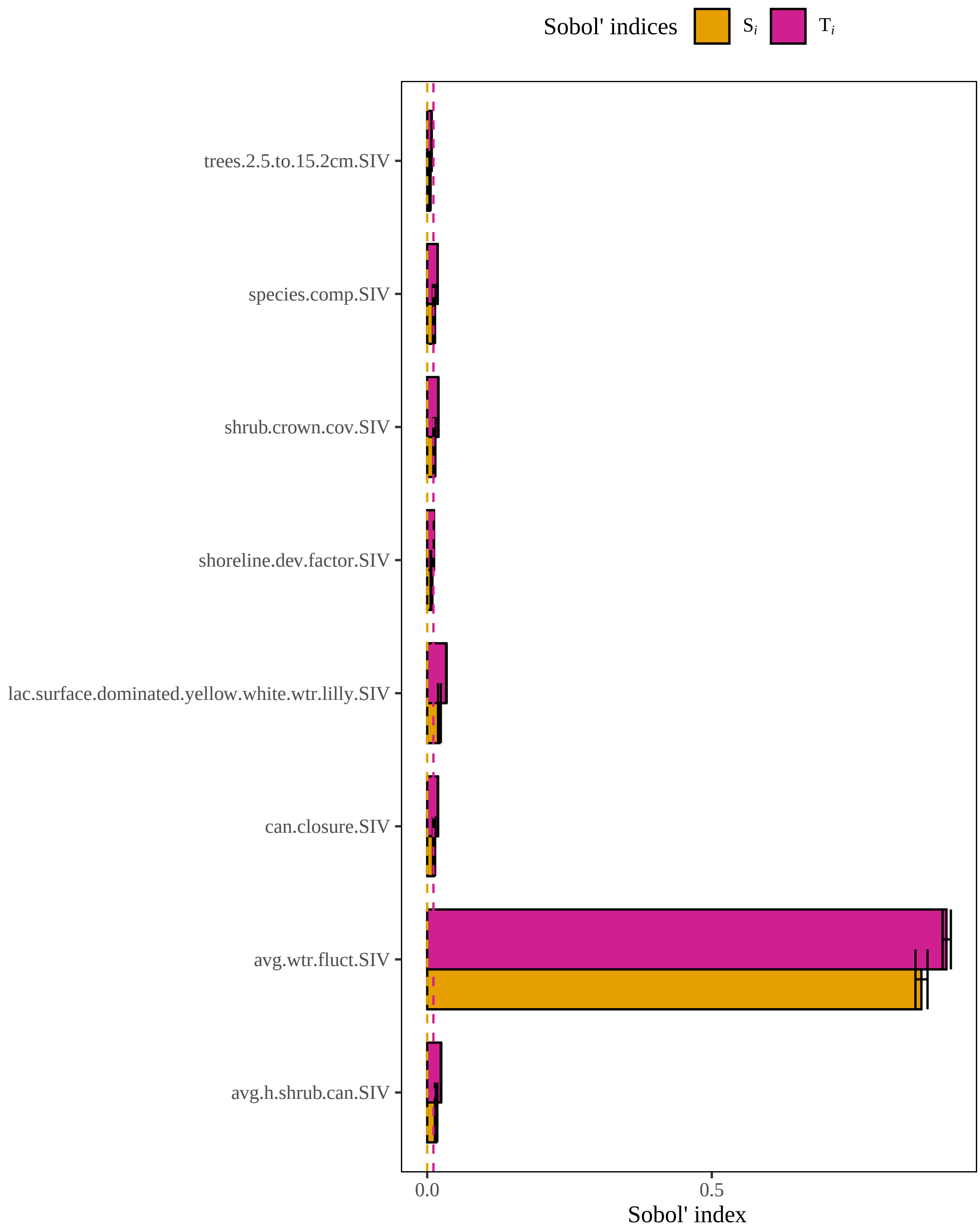

Figure 5. Results of a sensitivity analysis for the beaverLacAreaMtoe8ha model based on a geometric mean structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Limiting factor model

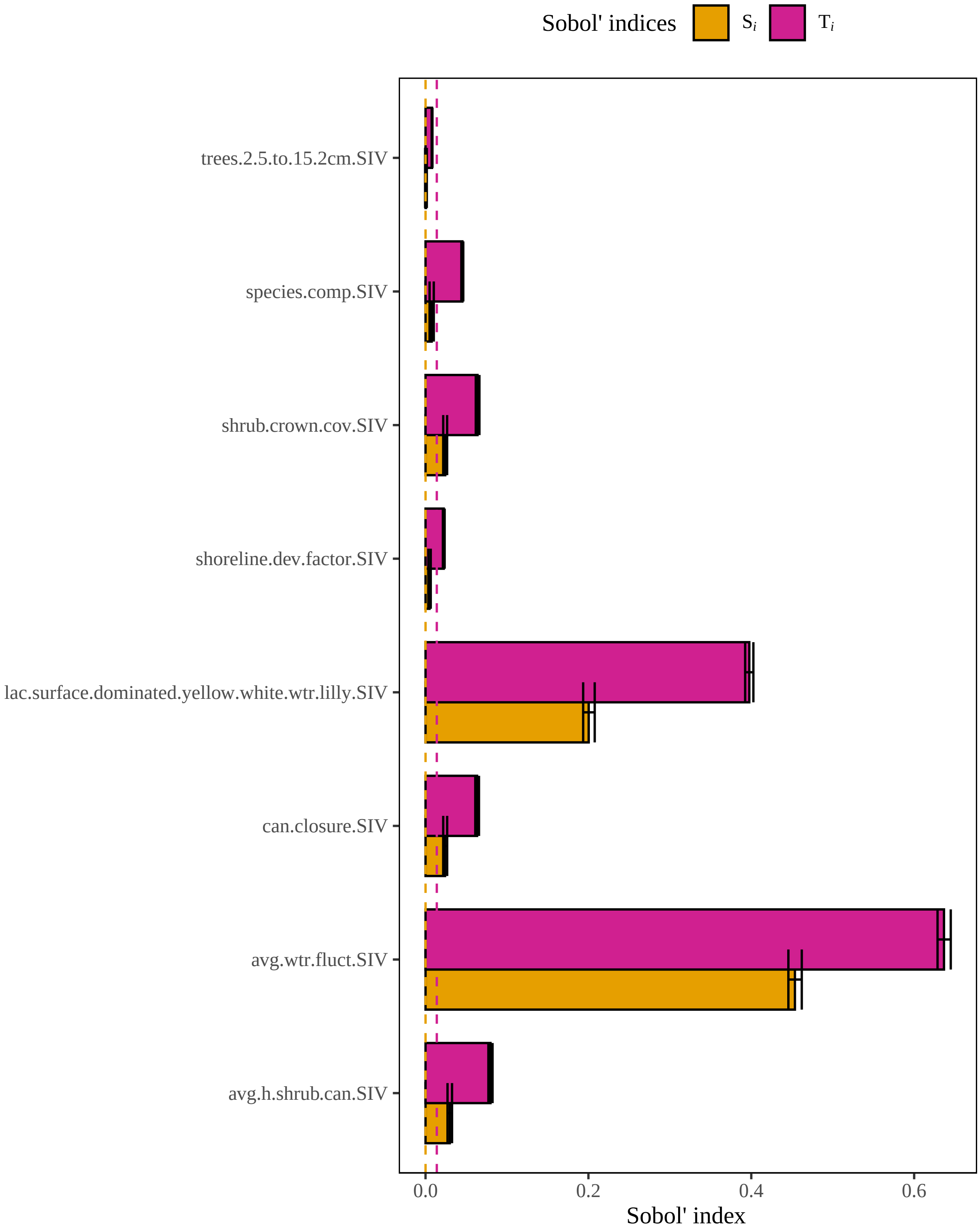

Figure 6. Results of a sensitivity analysis for the beaverLacAreaMtoe8ha model based on a limiting factor structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Multiplicative model

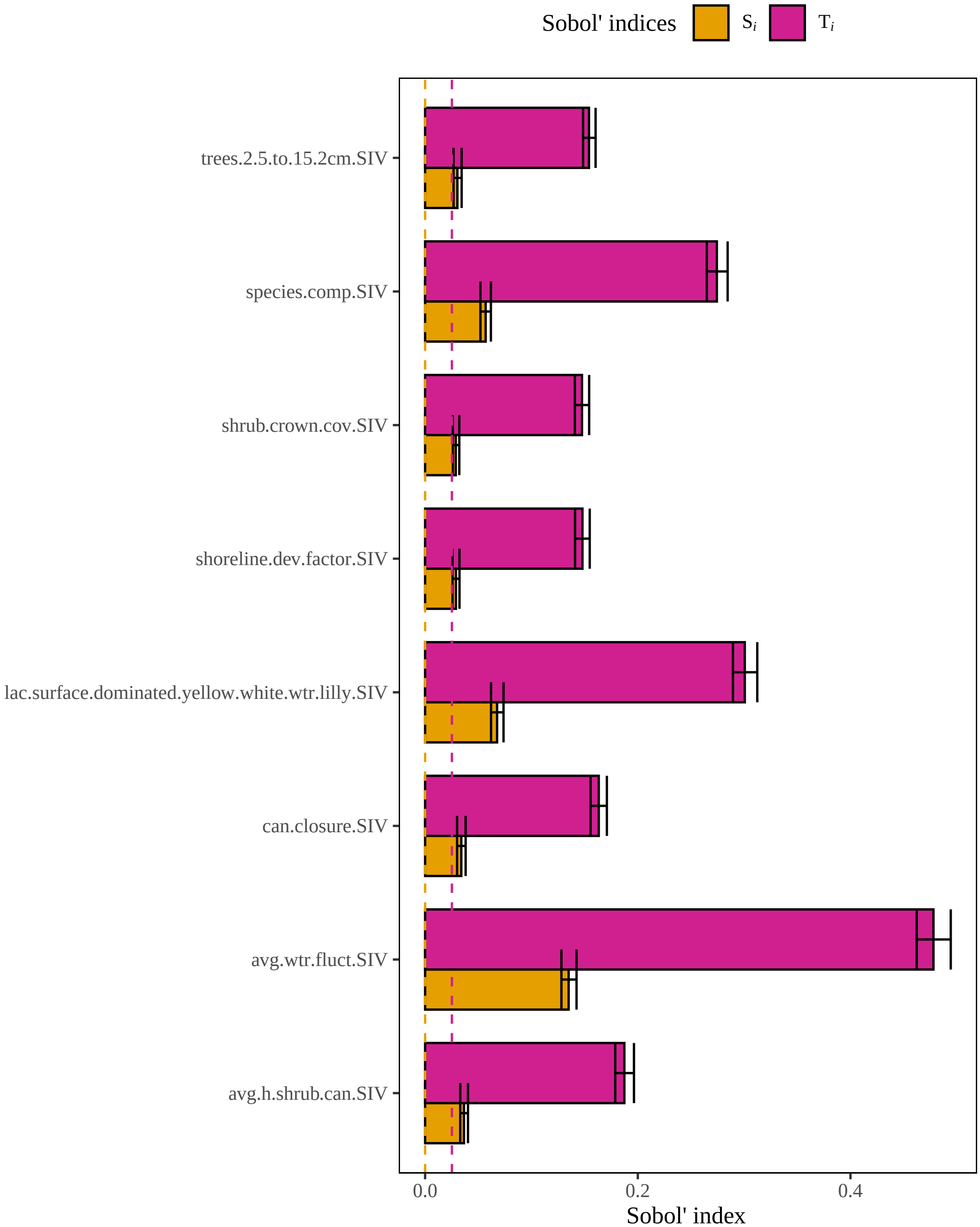

Figure 7. Results of a sensitivity analysis for the beaverLacAreaMtoe8ha model based on a multiplicative mean structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Summary of influential variables

Original model

In the **original beaverLacAreaMtoe8ha** model, **2 of 8** variables are influential.

Variable with the highest first order sensitivity: avg.wtr.fluct.SIV

Variable with the highest total order sensitivity: avg.wtr.fluct.SIV

Un-influential variables in original model:

```
can.closure.SIV
trees.2.5.to.15.2cm.SIV
shrub.crown.cov.SIV
avg.h.shrub.can.SIV
species.comp.SIV
lac.surface.dominated.yellow.white.wtr.lilly.SIV
```

Arithmetic mean model

In the **arithmetic mean beaverLacAreaMtoe8ha** model, **7 of 8** variables are influential.

Variable with the highest first order sensitivity: avg.wtr.fluct.SIV

Variable with the highest total order sensitivity: avg.wtr.fluct.SIV

Un-influential variables in arithmetic mean model:

```
lac.surface.dominated.yellow.white.wtr.lilly.SIV
```

Geometric mean model

In the **geometric mean beaverLacAreaMtoe8ha** model, **7 of 8** variables are influential.

Variable with the highest first order sensitivity: avg.wtr.fluct.SIV

Variable with the highest total order sensitivity: avg.wtr.fluct.SIV

Un-influential variables in geometric mean model:

```
trees.2.5.to.15.2cm.SIV
```

Limiting factor model

In the **limiting factor beaverLacAreaMtoe8ha** model, **7 of 8** variables are influential.

Variable with the highest first order sensitivity: avg.wtr.fluct.SIV

Variable with the highest total order sensitivity: avg.wtr.fluct.SIV

Un-influential variables in limiting factor mean model:

```
trees.2.5.to.15.2cm.SIV
```

Multiplicative model

In the **multiplicative beaverLacAreaMtoe8ha** model, **8 of 8** variables are influential.

Variable with the highest first order sensitivity: avg.wtr.fluct.SIV

Variable with the highest total order sensitivity: avg.wtr.fluct.SIV

Un-influential variables in multiplicative model:

None

References

1. Allen, AW. 1982. Habitat suitability index models: Beaver. U.S. Dept. Int., Fish Wildl. Serv. FWS/OBS-82/10.30. 20 pp
2. McKay S, D Hernandez-Abrams, and K Cushway. 2025. ecorest: conducts analyses informing ecosystem restoration decisions. R package version 2.0.1, https://CRAN.R-project.org/package=ecorest.
3. Puy, A, S Lo Piano, A Saltelli, and SA Levin. 2022. sensobol: an R package to compute variance based sensitivity indices. Journal of Statistical Software 102(5):1-37. doi: 10.18637/jss.v102.i05
4. Sarrazin, F., Pianosi, F., & Wagener, T. (2016). Global sensitivity analysis of environmental models: Convergence and validation. Environmental Modelling & Software, 79, 135–152. https://doi.org/10.1016/j.envsoft.2016.02.005